

FINAL PROJECT REPORT

Title : Application of Machine Learning for Market Segmentation

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1. INTRODUCTION

Market segmentation that is the strategic decomposition of a heterogeneous market into distinct, homogeneous groups that can each be targeted with tailored marketing strategies is one of the foundational concepts of modern marketing science. First formalised by Smith (1956), the principle has remained central to marketing theory and practice for nearly seven decades. What has changed dramatically in recent years is the methodology available to implement it. Traditional segmentation relied on demographic variables, survey data, and manual analysis approaches that are increasingly inadequate given the scale, velocity, and complexity of digital commerce data. This report examines how machine learning can be applied to implement market segmentation in a real-world commercial context, replacing intuition-based groupings with algorithmically derived, behaviourally grounded market segments.

The study applies unsupervised machine learning specifically K-Means and Agglomerative Hierarchical Clustering to the UCI Online Retail dataset, a real-world transactional record of 541,909 invoice line items from a UK-based e-commerce gift retailer covering December 2010 to December 2011. By engineering RFM (Recency, Frequency, Monetary) features from the transaction data and clustering the resulting customer profiles, the analysis produces four behavioural market segments, each characterised by a distinct purchasing pattern and each requiring a distinct marketing strategy. The entire analytical pipeline from raw data through to final segment labels and visualisations is implemented in a Jupyter notebook. This report describes the methodology, interprets the results, and connects them to market segmentation theory and applied marketing strategy.

1.1 Market Segmentation: Concept and Challenge

Market segmentation rests on a simple but powerful observation: not all customers in a market are alike. They differ in their needs, their purchase behaviour, their price sensitivity, and their long-term value to the firm. A business that treats all customers identically sending the same message, the same promotion, the same communication frequency to every account is in effect ignoring this heterogeneity. It is funding retention campaigns for accounts that will never return, offering discounts to customers who would have paid full price, and failing to recognise and reward the small group of accounts that generate the majority of its revenue.

The challenge in applying market segmentation to real commercial data is computational. Modern transactional databases contain millions of records, dozens of variables, and structural features, skewness, missing values, cancellations, outliers that make manual segmentation impractical. Machine learning addresses this challenge directly. Unsupervised clustering algorithms can process high-dimensional behavioural data at scale, identify latent groupings that would be invisible to manual analysis, and produce reproducible, quantitatively validated segment assignments. As Janardhanan and Muthalagu (2020) demonstrate, algorithmic segmentation can segment datasets of millions of records orders of magnitude faster than traditional methods, while Hernandez et al. (2025) and Sáez-Ortuño et al. (2023) confirm that machine learning-derived segments consistently outperform demographic groupings in predictive value for marketing personalisation.

1.2 Customer Segmentation as an Implementation of Market Segmentation

It is important to clarify the relationship between market segmentation and customer segmentation, as the two terms are sometimes used interchangeably but refer to related rather than identical concepts. Market segmentation is the strategic framework the principle that a market should be understood as a collection of distinct sub-groups rather than a uniform mass. Customer segmentation is one specific implementation of that framework: the application of behavioural data from existing customers to identify which sub-groups are present in the current customer base, and to target each group with differentiated strategy.

This study implements market segmentation through customer segmentation. The four segments identified by the clustering algorithm Champions, Loyal Customers, At-Risk Customers, and Lost / Inactive are behavioural market segments in the full strategic sense: each represents a distinct pattern of market participation, a different value proposition, and a different set of marketing levers. The fact that the data source is a customer transaction log rather than a broad market survey does not change the strategic nature of the output. As Smith (1956) himself noted, the essential criterion for a market segment is internal homogeneity and external heterogeneity a criterion that behaviourally defined clusters satisfy more rigorously than demographic groupings.

1.3 Structure of This Report

Section 2 documents the dataset and its quality challenges. Section 3 presents exploratory analysis of the business context. Section 4 describes the RFM feature engineering that operationalises the market segmentation framework. Section 5 presents the cluster count optimisation methodology. Section 6 documents the K-Means and Hierarchical Clustering results and model comparison. Section 7 profiles the four resulting market segments through five complementary visualisations. Section 8 translates each market segment into a concrete, differentiated marketing strategy. Section 9 concludes with limitations and directions for future work.

2. DATA UNDERSTANDING AND CLEANING

2.1 The UCI Online Retail Dataset

The UCI Online Retail dataset is a publicly available, real-world transactional record from a UK-based non-store online retailer specialising in wholesale gift items. It has been widely used in applied market segmentation research by Agus and Hasibuan (2025), Ardiansyah et al. (2026), and Chen et al. (2012) among others precisely because it presents the data quality challenges typical of real commercial transaction logs rather than pre-cleaned benchmarks. The raw dataset contains 541,909 rows and eight columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. Each row represents a single product line item on a single invoice, covering the period from 1 December 2010 to 9 December 2011 across 38 countries.

Three structural problems are immediately visible in the raw data that must be resolved before any market segmentation analysis can proceed. First, 135,080 rows, 24.9% of the dataset carry no CustomerID value. Without a customer identity, these rows cannot be attributed to any market participant and therefore cannot contribute to a segmentation model. Second, 8,905 invoices begin with the letter "C", the retailer's system notation for a cancellation or return. Including these would suppress revenue totals and distort recency calculations for any customer who returned a purchase after their last genuine order. Third, a small number of rows carry non-positive Quantity or UnitPrice values, representing internal stock adjustments rather than commercial transactions.

2.2 Cleaning Protocol

A four-step cleaning protocol is applied in the notebook. Step one removes all 135,080 rows with null CustomerID. Step two removes 8,905 cancelled invoices. Step three removes 40 rows with $\text{Quantity} \leq 0$ or $\text{UnitPrice} \leq 0$. Step four casts CustomerID from float to integer and computes Revenue as Quantity multiplied by UnitPrice. The effect of each step is shown in Table 1.

Table 1: Data Cleaning Steps and Row Counts

Step	Operation	Rows Remaining	Rows Removed
0	Raw dataset	541,909	—
1	Drop null CustomerID	406,829	135,080
2	Remove cancelled invoices (prefix "C")	397,924	8,905
3	Remove Qty ≤ 0 or Price ≤ 0	397,884	40
4	Compute Revenue = Qty $\times$ Price	397,884	0 (transformation)

After cleaning, the dataset retains 397,884 transaction rows representing 4,338 unique, identifiable market participants generating a total of £8,911,408 in revenue. These 4,338 customers form the population on which market segmentation is performed.

3. EXPLORATORY DATA ANALYSIS

Before building the market segmentation model, the notebook examines three aspects of the business context: revenue trends over time, geographic distribution, and transaction-level distributions. These analyses do not feed directly into the clustering algorithm, but they are essential for interpreting the market segments that emerge understanding the shape of the market in which the segments exist provides the commercial context that makes the segments meaningful.

3.1 Monthly Revenue Trend

Figure 1 plots total revenue by calendar month across the observation window. Revenue ranges from £523,000 in January 2011 to £1,163,000 in November 2011. The growth is not uniform: a post-Christmas trough in January is followed by recovery through Q1 and Q2 and a sharp pre-holiday peak in October and November. This seasonality is a structural feature of the market the retailer operates in a gift category with pronounced demand peaks around major calendar occasions.



Figure 1: Monthly Revenue, Dec 2010 – Dec 2011. The November peak (£1.16M) reflects pre-Christmas wholesale ordering. This seasonal structure shapes how Recency scores should be interpreted across market segments.

The seasonal pattern has a direct implication for market segmentation: two customers with identical purchasing loyalty may receive very different Recency scores depending purely on when within the year their most recent purchase fell. A customer who last ordered in November will appear highly engaged; a customer with identical annual purchasing frequency who last ordered in February will appear dormant. This limitation is acknowledged and discussed further in Section 9.

3.2 Geographic Market Distribution

Figure 2 shows revenue by country of customer registration for the top ten markets. The United Kingdom accounts for £7,308,391, 82.0% of total revenue confirming that this is predominantly a domestic UK wholesale market. The Netherlands (£285k), EIRE (£263k), and Germany (£221k) represent the largest international market segments by revenue, each contributing 2–3% of the total.

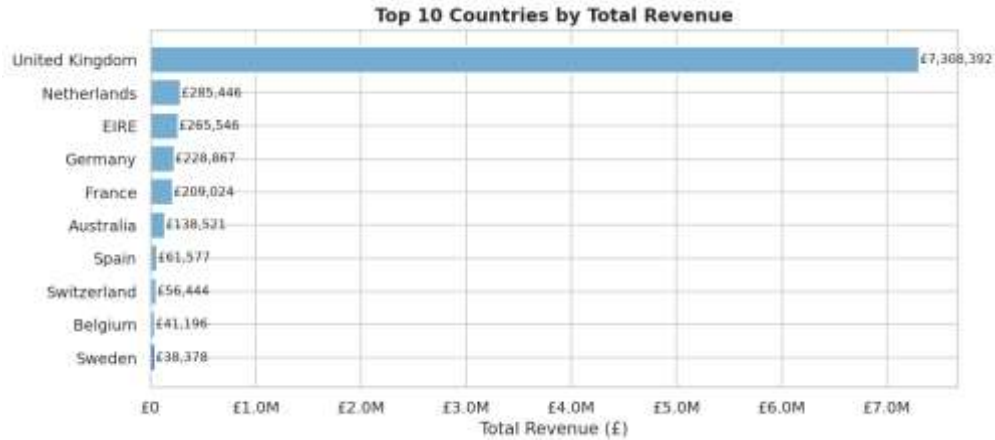


Figure 2: Top 10 Countries by Total Revenue. The UK accounts for 82.0% of total revenue. Geographic market concentration informs the scope within which behavioural segmentation is most commercially relevant.

Geography is deliberately excluded as a clustering feature. Market segmentation for commercial marketing purposes should be driven by behaviour purchasing patterns, engagement intensity, and spending levels rather than location. Two customers in different countries who exhibit identical behavioural profiles should belong to the same market segment and receive the same marketing strategy. Including Country as a feature would produce geographically rather than behaviourally defined segments, undermining the commercial usefulness of the model.

3.3 Transaction-Level Value Distribution

The median revenue per invoice line item is £9.90 with a mean of £22.39, indicating substantial right-skewness at the transaction level. This skewness propagates to the customer-level Monetary feature and is the primary motivation for the log-transformation applied during feature scaling.

4. RFM FEATURE ENGINEERING: OPERATIONALISING MARKET SEGMENTATION

The RFM framework, Recency, Frequency, Monetary is the feature engineering methodology that connects raw transaction data to the market segmentation objective. Originally developed in direct marketing (Hughes, 1994) and formalised for online retail contexts by Chen et al. (2012), RFM operationalises the three dimensions of customer behaviour that most directly predict future market participation: how recently a customer engaged with the market, how often they engage, and how much economic value they generate. These three dimensions define each customer's current position within the market and provide the basis for assigning them to a segment with others who share similar market behaviour.

The choice of RFM over other possible feature sets, demographic variables, product category preferences, geographic attributes, reflects a deliberate theoretical decision. Market segmentation based on RFM produces behavioural segments: groups defined by what customers actually do rather than who they demographically are. As Sáez-Ortuño et al. (2023) demonstrate, behavioural segmentation consistently outperforms demographic segmentation in predictive validity for marketing personalisation, because behaviour is more directly connected to future purchasing decisions than demographic identity.

4.1 Feature Computation

The 397,884 clean transaction rows are aggregated to one row per customer, computing three behavioural features against a snapshot date of 10 December 2011 one day after the last recorded transaction in the dataset. Recency is the number of days elapsed since the customer's most recent purchase. Lower Recency indicates more recently engaged market participants. Frequency is the count of distinct invoice numbers per customer, measuring purchase occasions rather than individual line items. Monetary is the sum of all Revenue values per customer across the observation period, expressed in British pounds.

Table 2: Descriptive Statistics of Raw RFM Features (n = 4,338 customers)

Statistic	Recency (days)	Frequency (invoices)	Monetary (£)
Mean	92.54	4.27	£2,054.27
Median	51.00	2.00	£674.48
Std Dev	100.01	7.70	£8,989.23
Min	1	1	£3.75
25th pct	18	1	£307.41
75th pct	142	5	£1,661.74
Max	374	209	£280,206.02

The divergence between median and mean across all three features most pronounced in Monetary (median £674 vs. mean £2,054) reflects the structural heterogeneity of the market. A small number of very high-value, very frequent customers dominate the upper tail of each distribution,

while the majority of customers cluster at low Frequency and moderate Recency values. This heterogeneity is precisely what market segmentation is designed to make visible and actionable.

4.2 RFM Distributions

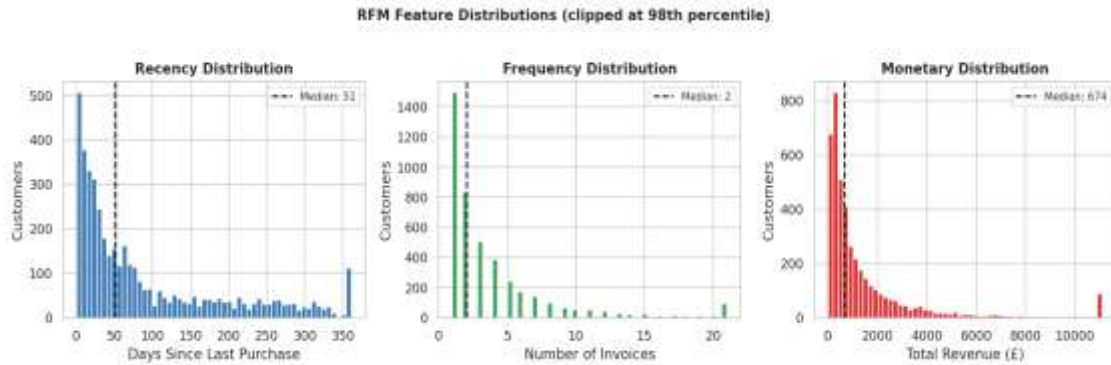


Figure 3: RFM Feature Distributions (raw values, clipped at 98th percentile). The right-skewness visible in all three panels confirms the structural market heterogeneity that motivates segmentation.

4.3 Feature Scaling

K-Means computes Euclidean distances in feature space. Without scaling, the Monetary feature (range: £3.75 – £280,206) would completely dominate the distance metric, making Recency and Frequency informationally irrelevant. A two-step procedure is applied: first, a $\log(1+x)$ transformation compresses the extreme right tails of all three features; second, StandardScaler normalises each feature to zero mean and unit variance. After scaling, all three behavioural dimensions contribute equally to the distance computation, and the resulting market segment boundaries reflect genuine differences in purchasing behaviour rather than differences in measurement scale.

5. DETERMINING THE OPTIMAL NUMBER OF MARKET SEGMENTS

Selecting the number of market segments K is the most consequential analytical decision in the segmentation pipeline. Setting K too low produces over-aggregated segments that mask important behavioural differences a single "low-value" segment might contain both recoverable at-risk customers and permanently churned accounts, requiring very different marketing strategies. Setting K too high produces artificially fine-grained segments that are statistically distinguishable but practically indistinguishable from a marketing execution standpoint, as no business can sustain twenty distinct campaign strategies simultaneously. This tension between statistical optimality and commercial practicality is a recurring theme in the market segmentation literature: Hernandez et al. (2025) found that while the mathematical optimum for their Chilean retail dataset was $K=12$, they constrained the final model to $K=4$ for commercial usability, a decision pattern consistent with this study.

Three independent internal validation metrics are computed for K values from 2 to 10. No external ground truth exists against which to validate this is the defining characteristic of unsupervised market segmentation, so internal metrics that measure cluster compactness and separation provide the only available quantitative evidence for segment quality.

5.1 Validation Metrics

Within-Cluster Sum of Squares (WCSS or Inertia) measures total within-segment variance. The Elbow Method identifies the K value at which the marginal gain from adding a segment diminishes markedly. The Silhouette Score measures how well each data point fits its assigned segment compared to the nearest alternative segment; higher values indicate better-defined market boundaries. The Davies-Bouldin Index measures the average ratio of within-segment scatter to between-segment distance; lower values indicate more compact, well-separated market segments.

Table 3: Market Segment Validation Metrics for $K = 2$ to 10

K	WCSS (Inertia)	WCSS Drop	Silhouette Score	Davies-Bouldin
2	6,481.2	—	0.4325	0.8921
3	4,867.8	1,613.4	0.3349	1.0480
4	3,938.5	929.3	0.3367	1.0117
5	3,296.0	642.5	0.3143	0.9895
6	2,855.0	441.0	0.3111	1.0134
7	2,548.9	306.1	0.3087	0.9722
8	2,336.8	212.1	0.2980	0.9940
9	2,155.6	181.2	0.2793	1.0183
10	1,999.9	155.7	0.2780	1.0192

4 = selected. Three independent metrics converge on $K = 4$ as the optimal number of market segments.

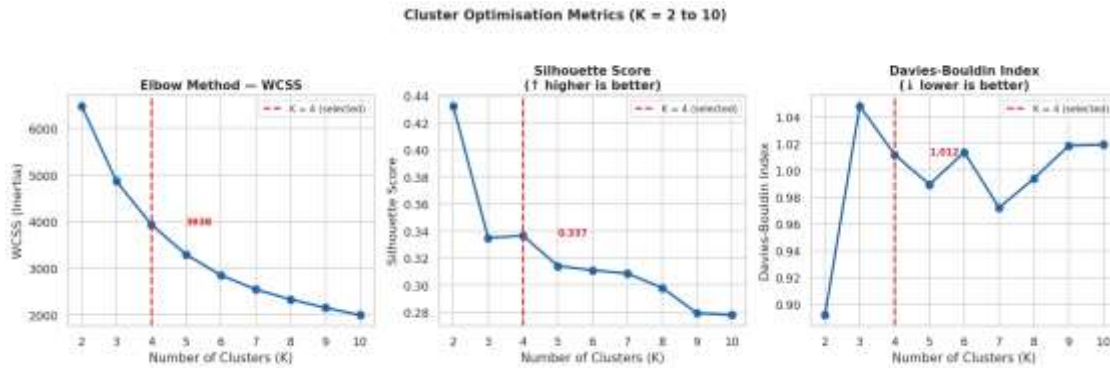


Figure 4: Market Segment Optimisation Metrics, K = 2 to 10. The elbow in WCSS at K=4.

The WCSS drop column makes the elbow quantitatively explicit. The marginal gain from adding a segment falls from 1,613 (K=2 to 3) to 929 (K=3 to 4) to 643 (K=4 to 5). K=4 is the last step where an additional segment reduces WCSS by more than 900 units. The Silhouette Score achieves its secondary local peak at K=4 (0.3367) after a primary maximum at K=2; the K=2 solution is statistically clean but commercially too coarse – a two-segment market structure offers no actionable differentiation beyond a simple "high-value versus low-value" split. The Davies-Bouldin Index reaches a local minimum at K=4 (1.0117). All three metrics independently identify four as the natural number of behavioural market segments in this dataset.

6. MACHINE LEARNING MODELS FOR MARKET SEGMENTATION

6.1 K-Means Clustering (K = 4)

K-Means is selected as the primary market segmentation algorithm for two reasons that align with the practical requirements of this analysis. First, it is computationally efficient at the scale of this dataset, Janardhanan and Muthalagu (2020) demonstrate that K-Means processes large retail datasets 97.8% faster than hierarchical alternatives, making it feasible for real-time or rolling segmentation applications. Second, its centroid-based structure produces interpretable segment profiles: each market segment is characterised by a centroid that represents the "typical" member of that group on all three RFM dimensions, providing a clear conceptual anchor for marketing strategy.

The model is configured with `n_clusters=4`, `init="k-means++"` (smart centroid initialisation that avoids worst-case convergence), `n_init=20` (twenty independent runs retaining the best result by inertia), and `random_state=42` for full reproducibility. The cluster labels produced by the algorithm are arbitrary integers 0–3; these are mapped to business-interpretable market segment names by ranking clusters on their mean Monetary value. Final model performance: Inertia = 3,938.5; Silhouette Score = 0.3367; Davies-Bouldin Index = 1.0117; Calinski-Harabasz Score = 3,329.2.

6.2 Hierarchical Clustering as Independent Validation

A fundamental challenge in market segmentation via unsupervised learning is that the same data can produce different results depending on the algorithm chosen and its initialisation. K-Means is sensitive to initial centroid placement and can, in theory, converge to locally rather than globally optimal solutions. To validate that the four-segment structure found by K-Means reflects genuine market heterogeneity rather than an initialisation artefact, Agglomerative Hierarchical Clustering with Ward linkage is applied as an independent second algorithm.

Ward linkage is chosen because it minimises total within-cluster variance at each merge step the same objective that K-Means optimises iteratively making the two algorithms directly comparable in intent while differing in procedure. The dendrogram (Figure 5) is constructed on a 300-customer random sample for visual clarity.

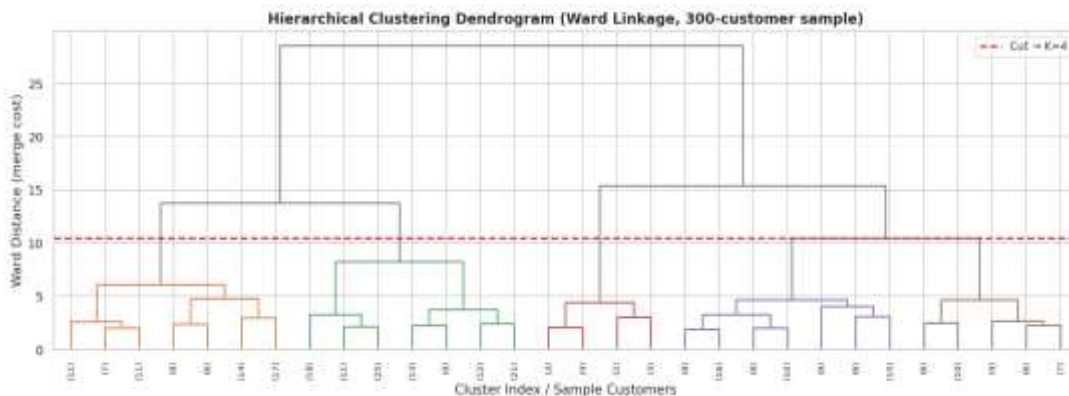


Figure 5: Hierarchical Clustering Dendrogram (Ward Linkage, 300-customer sample).

The dendrogram shows a clear horizontal gap above the fourth merge level the Ward distance at the K=4 cut point is substantially larger than the within-group merge distances immediately below it. This confirms that four natural groupings exist in the RFM data structure independently of any K-Means initialisation. Applying hierarchical clustering to the full 4,338-customer dataset with K=4 yields: Silhouette Score = 0.2385; Davies-Bouldin Index = 1.1200; Calinski-Harabasz Score = 2,615.1.

6.3 Model Comparison

Table 4: Algorithm Comparison for Market Segmentation (K = 4)

Metric	K-Means	Hierarchical (Ward)	Difference	Better
Silhouette Score ↑	0.3367	0.2385	+0.0982 (+41.2%)	K-Means
Davies-Bouldin Index ↓	1.0117	1.1200	−0.1083 (−9.7%)	K-Means
Calinski-Harabasz Score ↑	3,329.2	2,615.1	+714.1 (+27.3%)	K-Means

K-Means outperforms Hierarchical Clustering on all three validation metrics at K=4. The 41.2% Silhouette Score advantage indicates that K-Means produces market segment assignments where customers are substantially more similar to members of their own segment than to members of adjacent segments. This result is structurally expected: K-Means directly optimises cluster compactness at every iteration through centroid repositioning, whereas Ward linkage applies a greedy merge criterion that cannot revise earlier decisions. The hierarchical approach nonetheless serves its critical role as a structural validation, its independent confirmation of the four-segment market structure substantially increases confidence in the K-Means result. K-Means is selected as the operational market segmentation model.

7. MARKET SEGMENT PROFILES

This section presents the four market segments identified by the K-Means algorithm through five complementary visualisations. Each visualisation illuminates a different dimension of the market structure. The segment profiles are the primary output of the market segmentation analysis, they translate the algorithmic result into characterisations of distinct market sub-groups that can anchor differentiated commercial strategy.

Table 5: Market Segment Profiles , Behavioural Characteristics and Revenue Contribution

Segment	N	% of Market	Avg Recency	Avg Frequency	Avg Monetary	Total Revenue	Rev Share
Champions	705	16.3%	12 days	13.8 orders	£8,151	£5,746,587	64.5%
Loyal Customers	1,176	27.1%	70 days	4.1 orders	£1,820	£2,139,969	24.0%
At-Risk Customers	840	19.4%	18 days	2.1 orders	£544	£457,303	5.1%
Lost / Inactive	1,617	37.3%	183 days	1.3 orders	£351	£567,548	6.4%

Total market revenue: £8,911,408. Snapshot date: 10 December 2011.

7.1 Market Segment Size and Revenue Contribution

Figure 6 presents the distribution of customers across the four market segments (left) and the total revenue contribution of each segment (right). The most striking feature is the asymmetry between the Champions and Lost / Inactive segments. Champions represent 16.3% of the market by customer count but generate 64.5% of total revenue , £5,746,587 out of £8,911,408. This concentration is consistent with the Pareto principle documented across market segmentation studies in the RFM literature, and it has direct implications for marketing resource allocation.

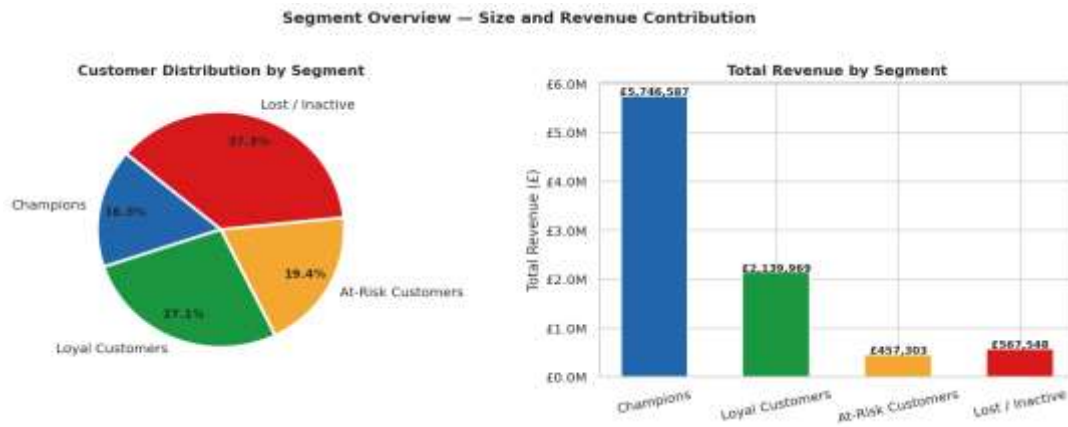


Figure 6: Market Segment Overview, Customer Distribution (left) and Revenue Contribution (right).

7.2 Behavioural Distributions Within Segments

Figure 7 shows boxplot distributions of each RFM dimension broken out by market segment. Outliers are suppressed at the 95th percentile and the mean is marked with a diamond symbol.

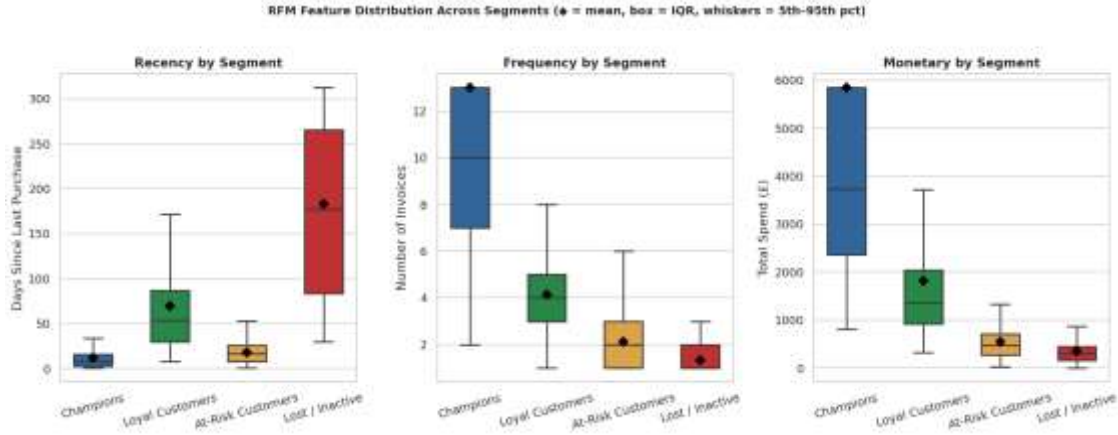


Figure 7: RFM Distributions by Market Segment.

The Recency panel shows Champions purchasing almost continuously (median approximately 10 days), while Lost / Inactive customers show a wide distribution centred above 150 days with a long tail approaching 374. The Frequency panel reveals the most dramatic between-segment contrast: Champions average 13.8 invoices per year, more than three times the Loyal Customers' 4.1. The Monetary panel confirms the revenue concentration: the interquartile range of the Champions distribution exceeds the entire value range of the At-Risk and Lost / Inactive segments, demonstrating that even the least active Champions generate more than most customers in those segments.

7.3 PCA Projection - Spatial Validation of Market Boundaries

Figure 8 projects the three-dimensional scaled RFM space onto its first two principal components, retaining 93.86% of the original variance (PC1: 75.10%, PC2: 18.76%). This visualisation serves as spatial validation of the market segment boundaries.

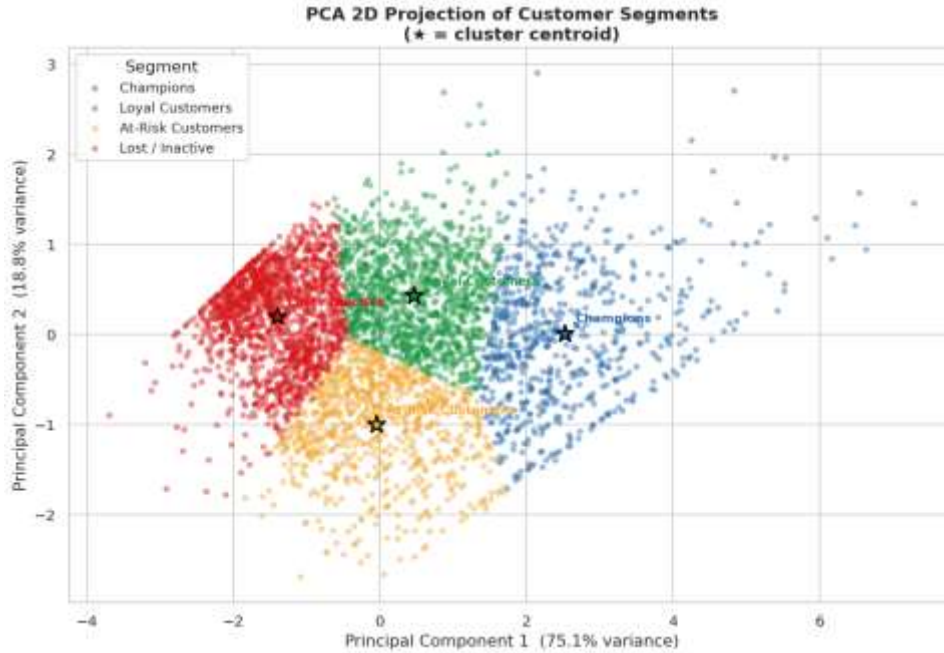


Figure 8: PCA 2D Projection of Market Segments (93.86% of variance retained). Stars mark segment centroids

The Champions centroid occupies the top-left of the projection, driven by high Monetary and Frequency values on PC1. The Lost / Inactive centroid is at the bottom-right, diagonally opposite reflecting low values across all three dimensions. Loyal Customers and At-Risk Customers occupy the central region but are separated on PC2, which captures primarily Recency variance. The clear spatial separation between segments in this nearly-complete projection of the original data confirms that the four market segments represent genuine structural divisions in the customer population, not arbitrary algorithmic partitions.

7.4 Segment Performance Heatmap

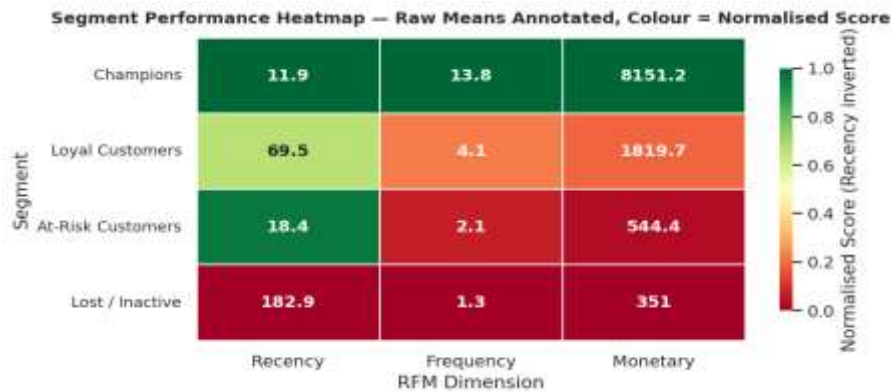


Figure 10: Market Segment Performance Heatmap. Colour reflects normalised performance (green = stronger; red = weaker; Recency inverted). Cell annotations show raw mean values, preserving commercial context alongside the normalised scoring.

The heatmap provides a consolidated summary of all segment-by-dimension combinations. Reading across the rows, Champions score green on all three dimensions. Loyal Customers show consistent yellow-green, indicating moderate but reliable market participation. At-Risk Customers present a distinctive pattern: a green Recency cell (18 days very recently engaged) alongside two red cells for Frequency and Monetary, confirming the recent-but-infrequent market participation profile visible in the radar. Lost / Inactive show uniformly red cells, with a Recency mean of 183 days confirming sustained disengagement from the market.

8. DIFFERENTIATED MARKETING STRATEGY BY MARKET SEGMENT

The commercial value of market segmentation lies in its translation into differentiated marketing strategy. If every segment receives the same campaign, the segmentation analysis has produced no practical improvement over the pre-existing one-size-fits-all approach. This section specifies a distinct marketing strategy for each of the four market segments, derived from and justified by the segment profiles documented in Section 7. The strategies are deliberately non-overlapping: treating Champions with the strategy designed for Lost / Inactive accounts or vice versa would reduce commercial value.

8.1 Champions

Seven hundred and five customers in the Champions segment generate £5,746,587 in annual revenue, 64.5% of the retailer's total. With an average recency of 12 days and average frequency of 13.8 invoices per year, these customers purchase almost continuously and at a level of intensity that indicates habitual, committed market participation. They do not need to be acquired or incentivised to return, they are already maximally engaged. The commercial risk is not failure to attract them but failure to retain them.

The marketing strategy for Champions must prioritise recognition and exclusivity over promotion and discounting. A generic promotional email sent to a customer generating £8,151 per year is a signal that the business does not recognise who they are and unrecognised customers defect. Champions should receive dedicated account management, priority access to new product lines before general release, and loyalty benefits scaled to their actual spending level. Discounting is counter-productive for this segment: these customers are already purchasing at full price and at maximum frequency. A promotional discount applied to a Champion does not generate incremental revenue it reduces margin on a sale that would have occurred regardless. Protecting the Champions segment from margin erosion while reinforcing their sense of being valued is the highest-priority marketing task the segmentation model identifies.

8.2 Loyal Customers

The 1,176 Loyal Customers contribute £2,139,969 (24.0% of revenue) with an average frequency of 4.1 invoices and an average recency of 70 days. They demonstrate consistent market participation but at a lower intensity than Champions. The gap between these two segments on Frequency, 4.1 invoices per year versus 13.8 represents the primary organic growth opportunity in the market structure. If even 25% of Loyal Customers could be moved to Champion-level purchase frequency, the revenue impact would exceed £3 million annually.

The strategy for Loyal Customers is to increase the number of purchase occasions without requiring margin concessions. Seasonal prompting communications aligned with Valentine's Day, Mother's Day, summer occasions, and back-to-school cycles gives customers who already have an established relationship with the retailer additional reasons to order outside their existing purchasing pattern. Product bundling and cross-category recommendations increase basket depth on each visit. A tiered loyalty programme, designed so that "the next level" feels proximate rather than distant, creates an ongoing incentive for increased purchasing frequency that compounds over time.

8.3 At-Risk Customers

The 840 At-Risk Customers are the most strategically urgent segment despite contributing only 5.1% of current revenue. Their average recency of 18 days, more recent than the Loyal Customers indicates that they are still behaviourally present in the market. But their average frequency of 2.1 invoices and average Monetary value of £544 reveal the risk: these are predominantly single or double-occasion buyers who have not established habitual market participation. The window for converting them from trial to loyalty is narrow if they do not return within 60 to 90 days of their most recent purchase, they will transition into the Lost / Inactive segment by the natural decay of market engagement.

The cost of retaining At-Risk Customers while they remain active is substantially lower than the cost of reactivating them after they have gone silent. The recommended intervention is a personalised follow-up campaign deployed within 30 days of each customer's last recorded purchase. The campaign should reference the customer's specific purchase history, suggest complementary products aligned with what they have already bought, and include a modest time-limited incentive calibrated to their historical average order value. The objective is not to offer a deep discount but to provide a reason and a moment for a second or third purchase. the repetition that converts occasional market participation into habitual engagement.

8.4 Lost / Inactive

The Lost / Inactive segment is the largest by customer count (1,617, 37.3%) but the smallest by revenue contribution (£567,548, 6.4%). Average recency is 183 days , six months since the most recent purchase and average frequency is 1.3 invoices, indicating that the large majority of these customers made a single purchase and never re-engaged with the market. This pattern is consistent with what the market segmentation literature describes as "one-and-done" acquisition: customers who entered the market through a specific occasion or promotion and found insufficient reason to return.

The appropriate marketing response is not equivalent investment to that applied to Champions or Loyal Customers. A full reactivation programme applied to 1,617 predominantly unresponsive accounts would misallocate resources that could generate higher returns in other segments. The recommended approach is a structured, low-cost automated reactivation sequence: two email communications over a 90-day window that acknowledge the gap in engagement, reference the customer's prior purchase, and offer a modest reactivation incentive. Customers who do not respond within 90 days should be reclassified as churned and removed from active marketing lists. This is resource discipline rather than abandonment it redirects the budget freed from unresponsive accounts into the Champions retention and Loyal Customers conversion programmes where return on marketing investment is demonstrably higher.

9. CONCLUSION

This study demonstrates that machine learning can be applied to produce statistically rigorous, commercially actionable market segmentation from real-world transactional data. The analytical pipeline cleaning, RFM feature engineering, log-scaling, cluster count optimisation using three independent metrics, and comparative evaluation of K-Means versus Hierarchical Clustering, identifies four distinct behavioural market segments in the UCI Online Retail dataset: Champions, Loyal Customers, At-Risk Customers, and Lost / Inactive. These four segments satisfy the foundational criterion of market segmentation as defined by Smith (1956): each is internally homogeneous with respect to purchasing behaviour and externally heterogeneous relative to the other segments.

The central finding which is that 16.3% of the market (Champions) generates 64.5% of total revenue is not merely a statistical observation. It is a structural insight about market composition that should govern how marketing resources are allocated. Every pound spent on uniform mass-market campaigns that treat Champions and Lost / Inactive accounts identically is a pound that fails to protect the revenue-generating relationships that sustain the business and simultaneously fails to convert the moderately engaged accounts that represent the highest organic growth potential. The four differentiated marketing strategies specified in Section 8 translate this structural insight into executable decisions: Protect Champions through recognition and exclusivity; Convert Loyal Customers through increased purchase occasion frequency; Rescue At-Risk Customers through timely, personalised intervention; and apply structured resource discipline to Lost / Inactive accounts.

Three limitations of this study should be acknowledged. First, the segmentation is a static snapshot of a single twelve-month observation window. Market segment membership is dynamic customers migrate between segments in response to service quality, competitive offers, and personal circumstances. A production deployment of this pipeline would require rolling monthly recomputation of RFM scores and segment reassignments to maintain accuracy. Second, RFM captures what customers buy but not why it cannot distinguish between a customer who went inactive due to dissatisfaction and one who went inactive due to a change in their own business needs. Incorporating customer service records, return behaviour, email engagement data, and product category preference data into the feature set would enable more diagnostic segment characterisation. Third, the study applies unsupervised clustering in the absence of outcome labels. A natural next step is to connect segment labels to subsequent purchasing behaviour, generate labelled outcome data, and train a supervised model, XGBoost, as recommended by Sáez-Ortuño et al. (2023), to predict future segment membership from current RFM profiles, enabling proactive rather than reactive segmentation strategy.

The broader contribution of this study is methodological: it demonstrates that the RFM-K-Means pipeline, applied with appropriate data cleaning, feature scaling, and multi-metric validation, produces market segments that are both statistically sound and commercially interpretable. The approach is reproducible, scalable, and directly connected to the foundational market segmentation theory articulated by Smith (1956), closing the gap between the strategic concept and its algorithmic implementation.

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